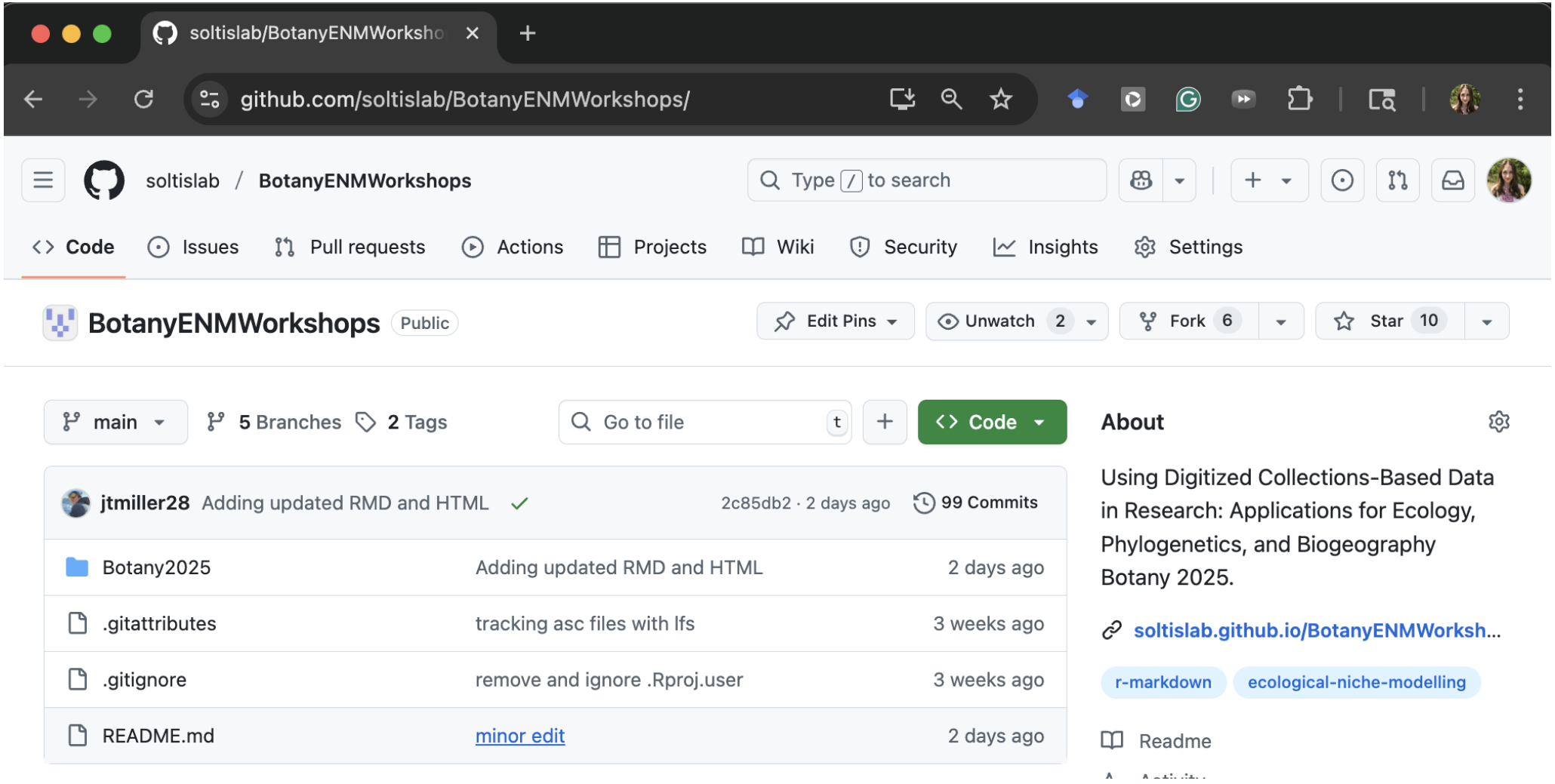


R-based Data Downloads

Shelly Gaynor
University of Florida

iDigBio
Integrated Digitized Biocollections

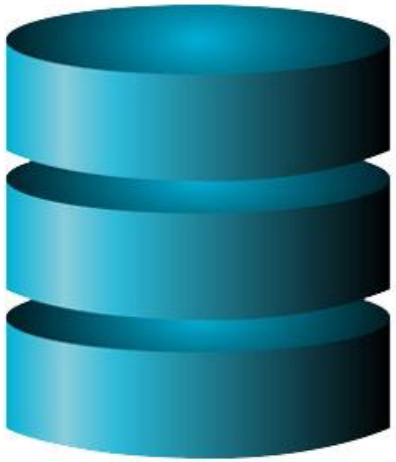
Github: soltislab/BotanyENMWorkshop/



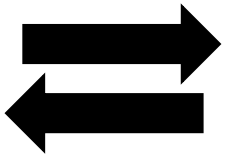
See the code: https://soltislab.github.io/BotanyENMWorkshops/

API = Application Programming Interface

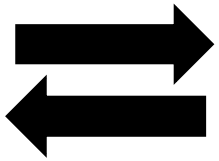
- Allows users to interact with a system



Database

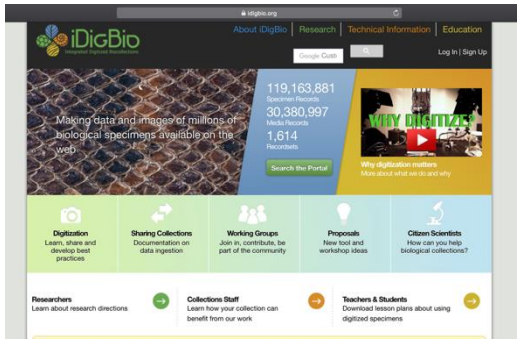


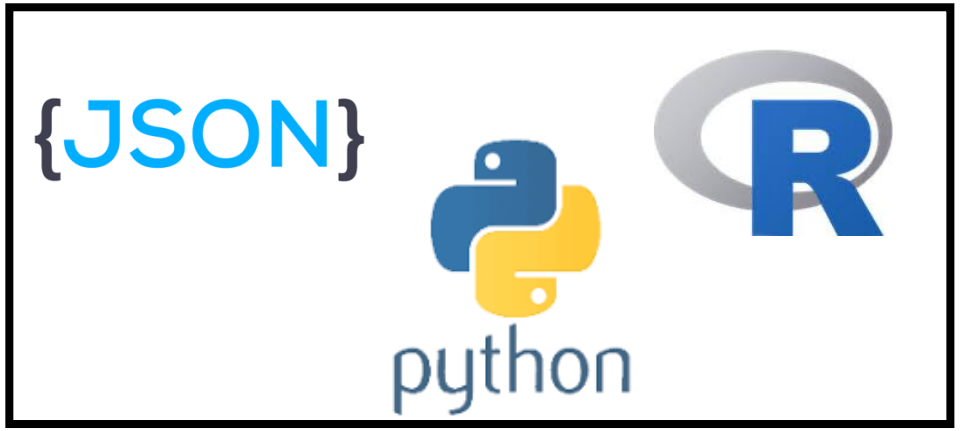
API



Interface

Web portal





idigbio.org

iDigBio
Integrated Digitized Biocollections

About iDigBio | Research | Technical Information | Education

ENHANCED BY

Log In | Sign Up

Making data and images of millions of biological specimens available on the web

121,428,342
Specimen Records
31,871,262
Media Records
1,621
Recordsets

Search the Portal



Why digitization matters
More about what we do and why

Digitization
Learn, share and develop best practices

Sharing Collections
Documentation on data ingestion

Working Groups
Join in, contribute, be part of the community

Proposals
New tool and workshop ideas

Citizen Scientists
How can you help biological collections?

Researchers
Learn about research directions

Collections Staff
Learn how your collection can benefit from our work

Teachers & Students
Download lesson plans about using digitized specimens

iDigBio API

- Multiple ways to access the API:

API Name	Info
Search API	ridigbio R package <100,000 records
Download API	>100,000 records
Record API	Single record
Media API	Single record



biodiversity-specimen-data/**specimen-data-use-case**

iDigBio API

- Multiple ways to access the API:

API Name	Info
Search API	ridigbio R package <100,000 records
Download API	>100,000 records
Record API	Single record
Media API	Single record



biodiversity-specimen-data/**specimen-data-use-case**

GBIF API

- Multiple ways to access the API:

API Name	Info
Registry API	Create, edit, update and search for information about datasets
Species API	Taxonomy API
Occurrence API	Record API
Maps API	Show maps of GBIF
News API	Search papers published using GBIF

GBIF Global Biodiversity Information Facility

https://www.gbif.org/developer/summary

Load Packages

```
library(ridigbio)
library(gatoRs)
library(leaflet)
```





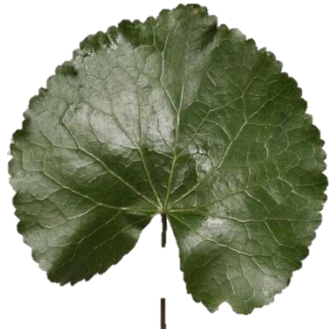


Downloading data using ridigbio

- First, we are searching for the species *Galax urceolata*
- Next, download occurrence records for the family Diapensiaceae

Search for the species Galax urceolata.

```
iDigBio_GU <- idig_search_records(rq=list(scientificname="Galax
 urceolata"))
```



Search for the family Diapensiaceae.

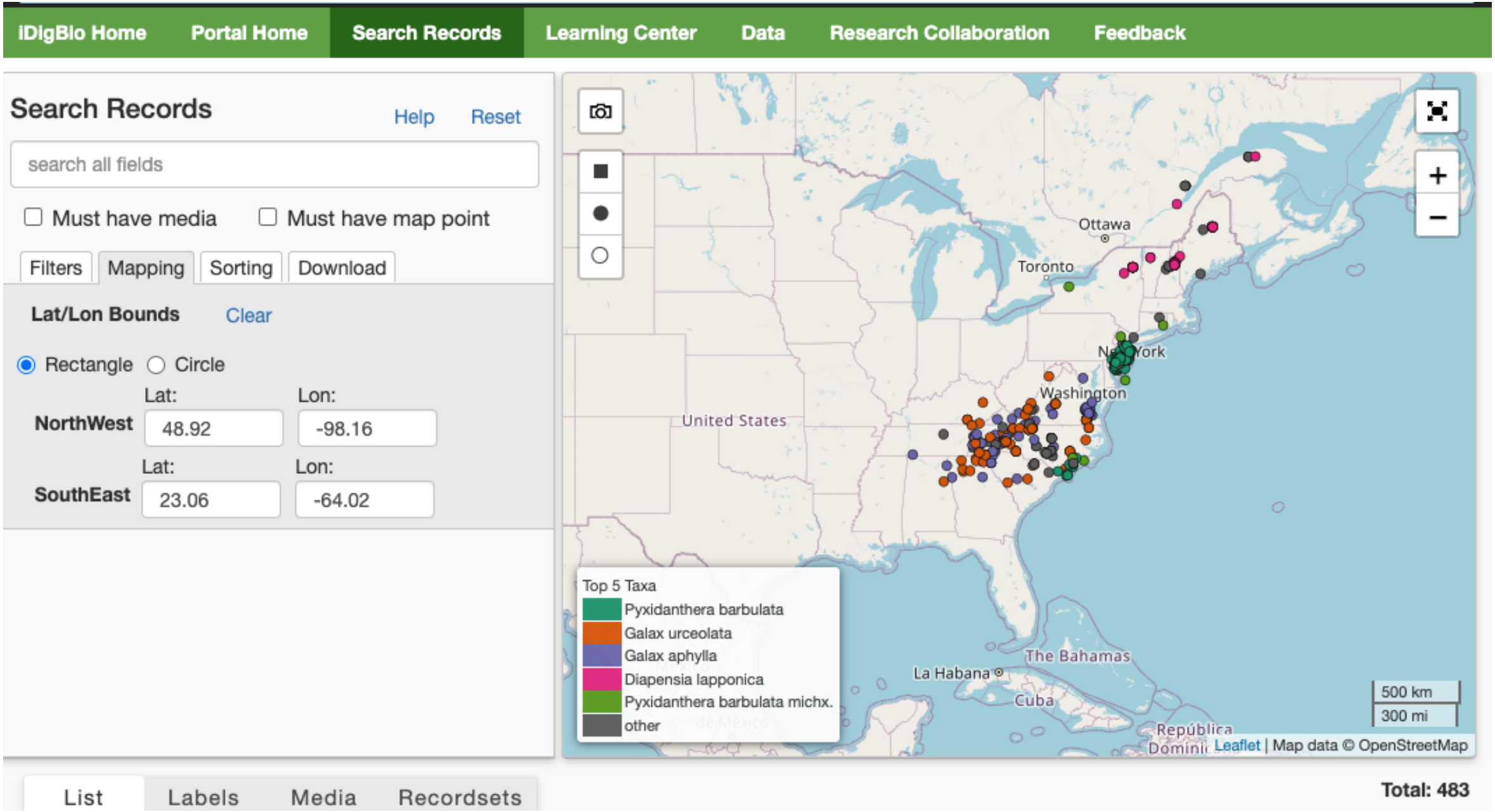
```
iDigBio_GU_family <- idig_search_records(rq=list(family="Diapen
siaceae"), limit=1000)
```

Records only in North America

```
rq_input <- list("scientificname"=list("type"="exists"),
                 "family"="Diapensiaceae",
                 geopoint=list(
                   type="geo_bounding_box",
                   top_left=list(lon = -98.16, lat = 48.92),
                   bottom_right=list(lon = -64.02, lat = 23.06)
                   )
                 )
```

Search using the input you just made

```
iDigBio_GU_family_USA <- idig_search_records(rq_input, limit=10
00)
```


iDigBio Home
Portal Home
Search Records
Learning Center
Data
Research Collaboration
Feedback
Search Records
Help
Reset
search all fields
Must have media
Must have map point
Filters
Mapping
Sorting
Download
Lat/Lon Bounds
Clear
Rectangle
Circle
Lat:
Lon:
NorthWest
48.92
-98.16
Lat:
Lon:
SouthEast
23.06
-64.02
Ottawa
Toronto
New York
Washington
United States
The Bahamas
La Habana
Cuba
Top 5 Taxa
Pyxidanthera barbulata
Galax urceolata
Galax aphylla
Diapensia lapponica
Pyxidanthera barbulata michx.
other
500 km
300 mi
Leaflet | Map data © OpenStreetMap
List
Labels
Media
Recordsets
Total: 483

Save as csv

```
write.csv(iDigBio_GU,
          file = "data/01_download/iDigBio_GU_2025_06_27.csv",
          row.names = FALSE)
write.csv(iDigBio_GU_family,
          file = "data/01_download/iDigBio_GU_family_2025_06_27.csv",
          row.names = FALSE)
```

Data download using gatoRs



Natalie Patten

- To pull data from GBIF and iDigBio for a set of synonyms
- Identifying synonyms:
  - Taxonomic Name Resolution Service
    - https://tnrs.biendata.org/
    - Used in soltislab/BotanyENMWorkshops 2020
  - R package taxize
    - 20 sources for synonyms
    - https://docs.ropensci.org/taxize/

iDigBio Integrated Digitized Biocollections

GBIF

gatoRs

Encyclopedia of Life

ITIS

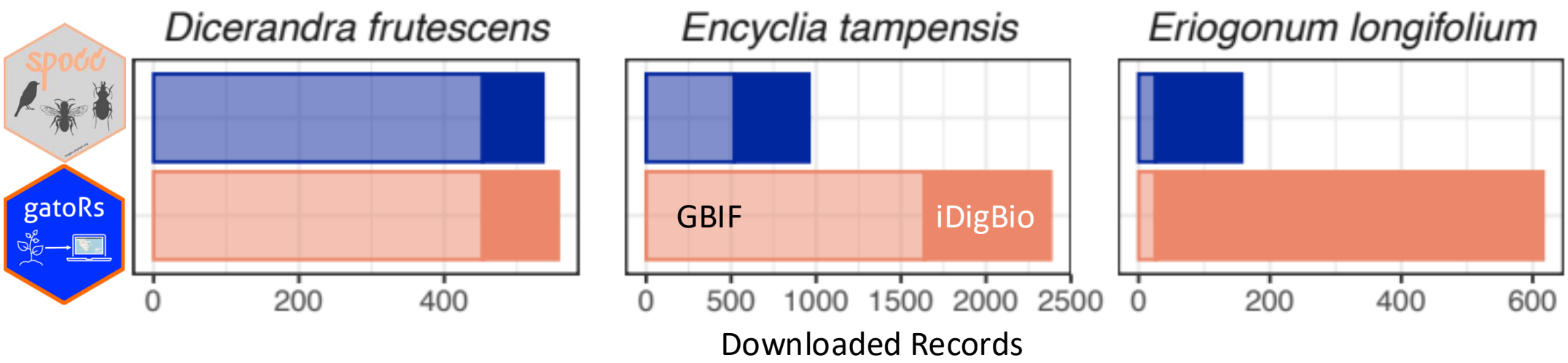
Tropicos

The Plant List

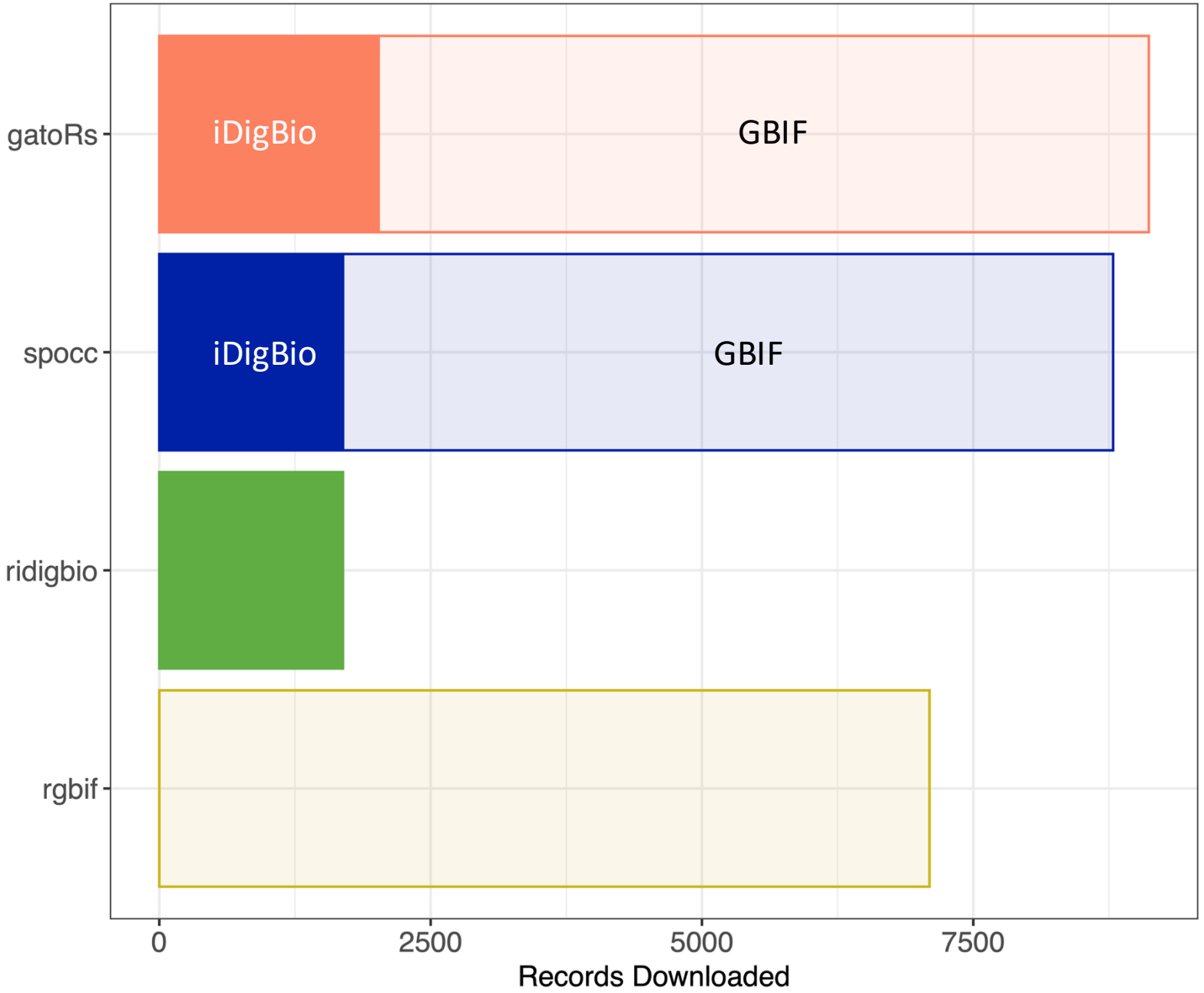
I·P·N·I

PLANTS

Why should I use gatoRs?



Patten et al. 2024. Applications in Plant Science.





Galax urceolata

https://mlgaynor.com/DEMO-Gaynor-2024/Gaynor-DEMO01.html

gatoRs Update - iDigBio

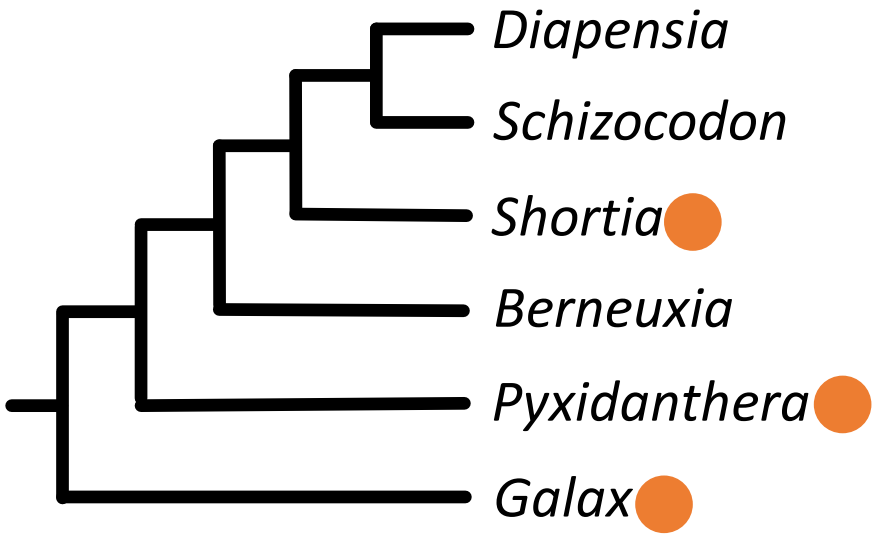
- In 2026, iDigBio moved to a GBIF hosted portal due to the end of their funding.
- A small subset of data providers did not move their recordsets over to the new hosted portal. This includes about 1.7 million records.
- In 2026, we modified gators_download() to pull these records from a GitHub-hosted archived download of these recordsets (downloaded on June 28, 2026).
- The full archived data set can be found on mgaynor1/OccurrenceArchive.
- Increases time to download.


Shortia galacifolia


Galax urceolata


Pyxidanthera barbulata


Pyxidanthera brevifolia



Diapensiaceae

Ericales

Data download using gator_download

Make synonym lists

Object

List of strings

```
Shortia_galacifolia <- c("Shortia galacifolia", "Sherwoodia galacifolia")
Galax_urceolata <- c("Galax urceolata", "Galax aphylla")
Pyxidanthera_barbulata <- c("Pyxidanthera barbulata","Pyxidanthera barbulat
a var. barbulata")
Pyxidanthera_brevifolia <- c("Pyxidanthera brevifolia", "Pyxidanthera barbu
lata var. brevifolia")
```

Data download using gators_download

```
gators_download(synonyms.list = Shortia_galacifolia,
                write.file = TRUE,
                filename = "data/download/raw/Shortia_galacifolia_raw_20230605.csv")
gators_download(synonyms.list = Galax_urceolata,
                write.file = TRUE,
                filename = "data/download/raw/Galax_urceolata_raw_20230605.csv")
gators_download(synonyms.list = Pyxidanthera_barbulata,
                write.file = TRUE,
                filename = "data/download/raw/Pyxidanthera_barbulata_raw_20230605.csv")
gators_download(synonyms.list = Pyxidanthera_brevifolia,
                write.file = TRUE,
                filename = "data/download/raw/Pyxidanthera_brevifolia_raw_20230605.csv")
```

Synonym list

Save csv file



Quick-look at downloaded files

Read in downloaded data frame

```
rawdf <- read.csv("data/download/raw/Shortia_galacifolia_raw_20230605.csv")
```

Inspect the data frame

What columns are included?

```
names(rawdf)
```

```
##  [1] "scientificName"              "genus"
##  [3] "specificEpithet"             "infraspecificEpithet"
##  [5] "ID"                          "occurrenceID"
##  [7] "basisOfRecord"               "eventDate"
##  [9] "year"                        "month"
## [11] "day"                         "institutionCode"
## [13] "recordedBy"                  "country"
## [15] "county"                      "stateProvince"
## [17] "locality"                    "latitude"
## [19] "longitude"                   "coordinateUncertaintyInMeters"
## [21] "informationWithheld"         "habitat"
## [23] "aggregator"
```

Where are these points?

The error message here indicates many points do not have long/lat values (more in 02).

```
leaflet(rawdf) %>%
  addMarkers(label = paste0(rawdf$longitude, ", ", rawdf$latitude)) %>%
  addTiles()
```

